Date

# LAB 8 Tasks BSA LABS

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Convert Infix → Postfix.

1. A + B * (C ^ D - E) ^ (F + G * H) - I

A + B * (C ^ D - E) ^ ( F + G * H) - I

| Scan | Stack | output | Description |
|---|---|---|---|
| A | | A | Operand → direct output |
| + | + | A | Stack empty push + |
| B | + | AB | operand output |
| * | +* | AB | * > + (precedence) → push |
| ( | +*( | AB | push ( |
| C | +*( | ABC | operand output |
| ^ | +*(^ | ABC | push ^ |
| D | +*(^ | ABCD | operand |
| - | +*(- | ABCD^ | pop ^ (higher precedence) push - |
| E | +*(- | ABCD^E | Operand |
| ) | +* | ABCD^E- | Pop till ( |
| ^ | +*^ | ABCD^E- | ^ > * → push |
| ( | +*^( | ABCD^E- | push ( |
| F | +*^( | ABCD^E-F | operand |
| + | +*^(+ | ABCD^E-F | push + |
| G | +*^(+ | ABCD^E-FG | operand |
| * | +*^(+* | ABCD^E-FG | * > + → push |
| H | +*^(+* | ABCD^E-FGH | operand |
| ) | +*^ | ABCD^E-FGH*+ | pop till ( |

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|---|---|---|---|
| - | - | ABCD^E-FGH*+^*+ | Pop ^ * , + (higher/equal precedence) then push - |
| I | - | ABCD^E-FGH*+^*+I | operand output |
| End | | ABCD^E-FGH*+^*+I- | pop remaining - |

Final postfix is ABCD^E-FGH*+^*+I-

2. A+ (B*C - (D/E^F)* G)*H.

| Scan | Stack | Output | Description |
|---|---|---|---|
| A | | A | operand → output |
| + | + | A | stack empty push + |
| ( | +( | A | push ( |
| B | +( | AB | operand |
| * | +(* | AB | push → * |
| C | +(* | ABC | operand |
| - | +(- | ABC* | pop * (higher) then push - |
| ( | +(-( | ABC* | push ( |
| D | +(-( | ABC*D | operand |
| / | +(-(/ | ABC*D | push / |
| E | +(-(/ | ABC*DE | operand |
| ^ | +(-(/^ | ABC*DE | push ^ (^ > /) |
| F | +(-(/^ | ABC*DEF | operand |
| ) | +(- | ABC*DEF^/ | pop ^ , / till ( |
| * | +(-* | ABC*DEF^/ | push * |
| G | +(-* | ABC*DEF^/G | operand |
| ) | + | ABC*DEF^/G*- | pop * , - till ( |
| * | +* | ABC*DEF^/G*- | push * |
| H | +* | ABC*DEF^/G*-H | operand. |

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|---|---|---|---|
| End | | ABC*DEF^/G*-H*- | pop remaining *, + |
| | Final postfix is | ABC*DEF^/G*-H*- | |

Convert Infix → Prefix.

1. A+(B*(C+(D*E)))

First reverse the expression

Reverse : (((E*D)+C)*B)+A

Now perform postfix of reversed expression

| Scan | Stack | Output | Description |
|---|---|---|---|
| ( | ( | | push ( |
| ( | (( | | push ( |
| ( | ((( | | push ( |
| E | ((( | E | operand output |
| * | (((* | E | push * |
| D | (((* | ED | operand output |
| ) | (( | ED* | pop * till ( |
| + | ((+ | ED* | push + |
| C | ((+ | ED*C | operand |
| ) | ( | ED*C+ | pop + |
| * | (* | ED*C+ | push * |
| B | (* | ED*C+B | operand |
| ) | | ED*C+B* | pop * |
| + | + | ED*C+B* | push + |
| A | + | ED*C+B*A | operand |
| End | | ED*C+B*A+ | pop remaining + |

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Reverse the output so $ED^*C + B^*A +$ will be

$+ A^* B + C^* DE$

Final prefix is $+ A^* B + C^* DE$

2. $A^* ( B + C^*(D - E)) + F/G$

Reverse : $G/F + ((E - D)^* C + B)^* A$

convert to postfix

| Scan | Stack | output | Description |
|---|---|---|---|
| G | | G | operand output |
| / | / | G | push / |
| F | / | GF | operand |
| + | + | GF/ | pop / (higher), push + |
| ( | +( | GF/ | push ( |
| ( | +(( | GF/ | push ( |
| E | +(( | GF/E | operand |
| - | +((- | GF/E | push - |
| D | +((- | GF/ED | operand |
| ) | +( | GF/ED- | pop - |
| * | +(* | GF/ED- | push * |
| C | +(* | GF/ED-C | operand |
| + | +(+ | GF/ED-C* | pop *, push + |
| B | +(+ | GF/ED-C*B | operand |
| ) | + | GF/ED-C*B+ | pop + |
| * | +* | GF/ED-C*B+ | push * |
| A | +* | GF/ED-C*B+A | operand |
| End | | GF/ED-C*B+A*+ | pop *, then + |

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Reverse the output so GF/ED - C* B + A* + will be

+* A + B* C - DE/FG.

Final prefix is + * A + B* C - DE/FG.

Q: Evaluate following expressions:

• 56 + 2 * 34 + - (Postfix) ↓

| Scan | Stack | Description. |
|---|---|---|
| 5 | 5 | push operand |
| 6 | 5 6 | push operand |
| + | 11 | pop 5, 6 → 5+6 = 11 push |
| 2 | 11 2 | |
| * | 22 | 11 * 2 = 22 |
| 3 | 22 3 | push |
| 4 | 22 3 4. | push |
| - | 22 -1 | 3-4 = -1 |
| + | 21 | 22 + (-1) = 21 |

Final answer = 21

• - * 4 5 + 2 3 (Prefix) ↓ right to left solved

| Scan | Stack | Description |
|---|---|---|
| 3 | 3 | push |
| 2 | 3 2 | push |
| + | 5 | 2+3 = 5 |

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|---|---|---|
| 5 | 5 5 | push |
| 4 | 5 5 4 | push |
| * | 5 20 | 4*5 = 20 |
| − | 15 | 20−5 = 15 |

final answer =15

- + * 2 34 (prefix) ↓ right to left solved

| Scan | Stack | Description |
|---|---|---|
| 4 | 4 | push |
| 3 | 4 3 | push |
| 2 | 4 3 2 | push |
| * | 4 6 | 2*3 = 6 |
| + | 10 | 6+4 = 10 |

final answer =10

- 9 3 * 4 2 / − (postfix) ↓

| Scan | Stack | Description |
|---|---|---|
| 9 | 9 | push |
| 3 | 9 3 | push |
| * | 27 | 9*3 = 27 |
| 4 | 27 4 | push |
| 2 | 27 4 2 | push |
| / | 27 2 | 4/2 = 2 |
| − | 25 | 27−2 = 25 |

final answer = 25